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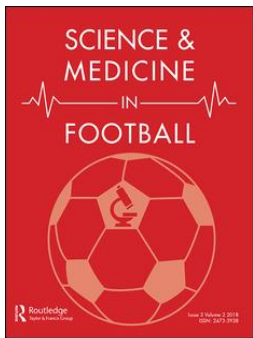


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To link to this article: <https://doi.org/10.1080/24733938.2020.1761558>



Accepted author version posted online: 24 Apr 2020.
Published online: 05 May 2020.



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Effects of training intensity and environmental condition on the hydration status of elite football players

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ABSTRACT

Objective: To examine the effects of training intensity and environmental condition on the hydration status of Elite football players. **Methods:** Eleven elite football players completed three training sessions of varying intensity in cool (12°C) and warm (23°C) environments. Training demands was measured by Global Positioning System, sweat rate and sweat sodium concentration were measured using dermal patches and body mass change. **Results:** Warm condition increased sweat rate (0.9 ± 0.3 vs 1.7 ± 0.3 L.h⁻¹, $P < 0.001$), fluid intake (0.7 ± 0.1 vs 1.5 ± 0.2 L, $P < 0.001$), total sodium loss (1405 ± 340 vs 2946 ± 958 mg, $P < 0.001$) and total sweat loss (1.4 ± 0.5 vs 2.5 ± 0.4 L, $P < 0.001$) compared to cool. Training intensity increased sweat sodium concentration (16.1 ± 6.6 vs 54.6 ± 22.2 mmol.L⁻¹, $P < 0.001$) and sodium loss (779 ± 231 vs 1405 ± 340 mg) in both environmental conditions. Total sweat loss and sodium loss were positively correlated with total distance covered ($r = 0.48$, $P = 0.005$ and $r = 0.4$, $P = 0.023$, respectively), meanwhile sodium loss was also positively correlated with the total number of high-intensity efforts ($r = 0.35$, $P = 0.045$). **Conclusions:** The results show that training load and environmental condition have a major impact on the hydration status of elite football players, hence hydration strategies should be developed accordingly.

ARTICLE HISTORY

Accepted 21 April 2020

KEYWORDS

Sodium; electrolytes; dehydration; soccer

Introduction

Dehydration during exercise occurs from the need of the body to maintain its temperature close to the normal value of 37°C, via thermoregulatory mechanisms involving sweat loss. When sweat loss is not matched by liquid consumption, body dehydration occurs, which at certain magnitudes impairs cardiovascular, thermoregulatory and mental functions (Armstrong et al. 1985; Sawka 1992; Broad et al. 1996; Nuccio et al. 2017). An increase in cardiovascular strain (i.e. heart rate drift and decreased stroke volume to maintain a given cardiac output) due to a decrease in plasma volume is classically observed with hypohydration. As a direct consequence and when fluid intake does not compensate for the body water losses, blood supply to active muscle is reduced (Gonzalez-Alonso et al. 1999), skeletal muscle metabolism is altered (i.e. increased glycogenolysis and carbohydrate oxidation), and thermoregulation is impaired (Logan-Sprenger et al. 2013; Cheuvront and Kenefick 2014). Castro-Sepulveda et al. (2018) recently added that commencing a match hypohydrated increased the cortisol response after the match, a physiological stress indicator. Furthermore, when body water loss exceeds 2–3% of body mass, it may negatively affect exercise performance (Barr 1999; Davis et al. 2015; Nuccio et al. 2017). In a football-specific study, McGregor et al. (1999) showed that 2.5% dehydration increased the rating of perceived exertion and decreased dribbling skills by 5% following the 90 min Loughborough shuttle running test. Similarly, performance at the Yo-Yo intermittent recovery test was impaired after a football match inducing a 2.4% dehydration compared

to 0.7% dehydration (Edwards et al. 2007). Although dehydration may impair football specific performance, it must be considered that players may be more or less sensitive to dehydration and some studies have reported no or limited effect of dehydration on football performance (Owen et al. 2013). The sweating process involves not only water losses, but also electrolyte losses, which can be detrimental to exercise performance if not replaced accordingly (Latzka and Montain 1999). For instance, electrolyte disturbances, particularly sodium loss, have been partly associated with exercise-induced muscle cramps, a condition commonly faced by team-sport athletes (Maughan et al. 2004; Stofan et al. 2005; Horswill et al. 2009). Both sweat loss and electrolyte loss are exacerbated in hot compared to temperate conditions (Mohr et al. 2012). The combination of dehydration and heat stress may therefore dramatically alter football performance.

Professional football is an intermittent team sport that is characterised by low-intensity activities that account for more than 70% of the duration of the game, interspersed with 150–250 short but intense actions (Bangsbo et al. 2007). The increase in metabolic rate sustained by football players during training and match-play causes sweat loss. Dehydration (i.e. decrease in body mass) is commonly observed in football and has been reported in many field studies conducted both in cool (Maughan et al. 2007) and warm environments (Shirreffs et al. 2005; Duffield et al. 2012). A high inter-individual variation in sweat loss, body mass loss and sodium loss is generally reported (Maughan et al. 2004, 2005; Pickering et al. 2011; Duffield et al. 2012). Multiple factors may

influence sweat loss and sodium loss, including training intensity and duration, weather conditions and intrinsic differences between players (Godek et al. 2010). Fluid intake during training also shows high inter-individual variations, but more importantly often does not match sweat loss, indicating that soccer players may be at risk of hypohydration during both training and competition (Maughan et al. 2004; Duffield et al. 2012).

Limited field data that illustrates how training intensity and environment influence the hydration status in elite football players is reported in the literature. Previous work conducted by Duffield et al. (2012) showed that training load (i.e. distance covered and intensity) was the main driver for sweat and sodium loss in professional football players. However data collection in this study took place in a warm environment (26.9°C), limiting applications to colder conditions. Previously, Shirreffs et al. (2005), recorded sweat loss and fluid intake of professional football players during a training session performed in a warm environment only (32°C). In the latter study, players presented a 1.6% body mass loss despite *ad libitum* drinking made possible. Training intensity was not controlled and there was no comparison with a temperate condition. Within this context, it appears that additional field-based data is required to understand the influence of the environment and training characteristics on the hydration status of football players. Ultimately bespoke hydration strategies could be proposed. The aim of the present investigation was, therefore, to assess changes in body mass, sweat rate, fluid intake, sodium loss and sweat sodium concentration in elite soccer players during low and high intensity pitch-based training sessions conducted in cool and warm environmental conditions. It was hypothesised that both training load and environmental conditions would influence sweat rate and sodium loss.

Methods

Participants

Eleven players of the first-team squad of an English Championship League football team took part in this study. This represented the entire first team squad except for the players who were unable to take part in any of the three training sessions included in this study due to illnesses, injuries, absence on international duty or transfers to other clubs during the timeframe of the study. Seventeen players took part in the first testing session, eighteen in the second one and twenty in the third one, but only eleven were able to attend the three training sessions. Hence, all the data presented in this article will refer only to the 11 players who undertook the three training sessions. The mean $\pm$ SD characteristics of the participants were: age: 25 ± 5 years, height: 1.8 ± 0.05 m and weight: 79.6 ± 8.8 kg. The players gave their oral and written informed consent to participate in the study. The protocol was in conformity with the Declaration of Helsinki (last modified in 2013) and was approved by the Research Ethics Committee of Liverpool John Moores University.

Study design

Testing and data collection were conducted during three training sessions. Two training sessions took place in a cool environment (Cool) during the winter season, whereas one took place in a warm environment (Warm) during the summer (Table 1). The intensity of training varied from low (LI) to high (HI) according to the manager's directives (Table 2) and was measured through the Global Positioning System (GPS). The training sessions were not modified for this study and were representative of sessions normally carried out by the team at the habitual days of the week and times of the season. They consisted of a warm-up, interval running, ball drills and football games. All training sessions took place between 10:30 am and 12:00 noon and lasted 86.6 ± 4.7 min (80 min, 90 min and 90 min for session 1, 2 and 3, respectively), with all the players following the same training protocol.

Study protocol

Training sessions

The HI and LI training sessions included both technical and physical exercise components, which were completed by the players at a higher and lower intensity, respectively. These definitions were based on both coach-led directives on the difficulty of the training sessions (i.e. hard or easy) and data collected through the GPS system (Catapult Vector, Catapult, Australia) worn by the players, more specifically the total distance covered and the relative intensity of movement. Distance covered at high intensity was defined as total distance covered at a speed $> 5.7 \text{ m.s}^{-1}$; high-intensity efforts were defined as the frequency of efforts $> 5.7 \text{ m.s}^{-1}$; intense accelerations were defined as the number of accelerations $> 2.5 \text{ m.s}^{-2}$.

Sweat loss and changes in body mass

The players were weighted wearing only underpants before and after the training sessions using the same scale (Seca 875, Hamburg, Germany) they used at the club. Sweat loss was calculated based on changes in body mass after correction for

Table 1. Weather conditions during the three training sessions of testing.

	Temperature (°C)	Relative Humidity (%)
1 st Training Session (Low-intensity, cool environment)	11	90
2 nd Training Session (High-intensity, cool environment)	12	81
3 rd Training Session (High-intensity, warm environment)	23	63

Table 2. Characteristics of training sessions, including total distance covered, distance covered at high intensity, number of high-intensity efforts, sprint distance and number of sprint efforts during the low-intensity (LI) and high-intensity (HI) training sessions. "denotes significance between the LI Cool testing session and the HI Cool testing session, * denotes significance between the LI Cool testing session and the HI Warm testing session, where $P < 0.05$.

	LI Cool	HI Cool	HI Warm
Total Distance Covered (m)	6008 ± 390	7780 ± 1629 "	7459 ± 567 *
Distance Covered at HI (m)	141 ± 62	291 ± 122 "	225 ± 103
Number of HI Efforts	12 ± 5	30 ± 15 "	26 ± 13 *
Number of Intense Accelerations	19 ± 6	23 ± 8	24 ± 8

fluid intake. Players were asked to urinate and defecate prior to the pre-training measurements. No player urinated during any of the three training sessions. Each player was provided with one 800 ml bottle of water and one 800 ml bottle of in-house made sport drink (400 mg Na⁺, 0 kcal, 0 g carbohydrates) to drink during the training sessions. Bottles were weighted using a kitchen scale (Kern, type C, Balingen, Germany) before they were provided to the players and after the training sessions to assess fluid intake. Additional bottles were available for the players during training if required. Habitual hydration practices during training sessions were maintained, with no extra encouragement nor recommendation given to the players regarding fluid intake.

Sodium losses

Sweat samples were collected during the three training sessions from three skin sites (back, forearm and thigh) by using absorbent sterile patches (10 x 10 cm, Health Care, Neuss, Germany) applied directly onto the surface of the skin. Prior to application, the target sites were cleaned with deionized water and dried. The patches were applied before the training sessions and remained in place throughout the sessions. All patches were placed on the right side of the body. At the end of the session, the patches were carefully separated from the adhesive tape using sterile tweezers before being inserted into the tube of a single use syringe. A sweat sample of approximately 5 mL was subsequently extracted by squeezing each wet patch into the syringe. All three sweat samples were then immediately analysed for sodium concentration by using a portable sodium meter (Horiba LAQUAtwin, Kyoto, Japan) previously tested for validity and reliability, and the average of the three results was used for determining the sweat sodium concentration of each player (Baker et al. 2014). The device was calibrated with known Na⁺ concentrations before utilisation and sample analysis was conducted according to the recommendations made by Goulet et al. (2012). Total sodium losses during the training sessions was then calculated according to the recommendations of the Gatorade Sport Science Institute (Barrington, IL, USA) based on the prediction equations developed by Baker et al. (2009), where:

- Total sweat [Na⁺] (mmol/L) = 0.57 (forearm sweat Na⁺) + 11.05
- Total sweat Na⁺ Loss (mmol) = Total Sweat Loss * Total Sweat [Na⁺]

Total sweat Na⁺ loss was then converted from mmol to mg using the molar mass of Na⁺ (22.99 mg/mmol) in the following equation:

- Total sweat Na⁺ loss (mg) = Total sweat Na⁺ loss * 22.99 mg/mmol * 1 g/1000 mg

Statistical analysis

A one-way analysis of variance (ANOVA) with repeated measures was used to determine differences in sweat rates, fluid intake, sweat sodium concentrations, sodium loss and changes in body mass between the three training sessions. Post-hoc

Bonferroni analysis was completed for all measurements, with significance set to $P < 0.05$. Effect sizes were also calculated using Cohen's d values, with values ≤ 0.2 , 0.5, 0.8, and > 0.8 considered as trivial, small, medium and large effects, respectively. Pearson's correlation coefficients were calculated to determine the relationship between selected variables of exercise intensity and sweat loss and sodium loss. The data presented in text, tables and figures are expressed as mean $\pm$ standard deviation, unless otherwise indicated. Statistical analysis was completed using SPSS for Windows (version 25, SPSS Inc, Chicago, IL).

Results

Training sessions' characteristics

Training sessions' characteristics are presented in Table 2. Total distance covered and distance covered at high intensity during LI Cool were 23% ($P = 0.005$) and 52% ($P = 0.006$) lower than during HI Cool, respectively. Total distance covered was also lower during LI Cool (19%, $P < 0.001$) than during HI Warm. Distance covered at high intensity during LI Cool tended to be lower (37%) than during HI Warm but it was not statistically significant ($P = 0.152$). There was no statistical difference for total distance covered ($P = 1.000$) and distance covered at high intensity ($P = 0.158$) between the two HI training sessions completed in different environments. The number of HI efforts was higher during both HI training sessions (117%, $P = 0.024$ and 150%, $P = 0.005$, in warm and cool environments, respectively) compared to LI training session, however there was no difference between the two HI training sessions for the number of HI efforts completed ($P = 0.853$). The number of accelerations was not different across all training sessions regardless of training intensity or environment ($P = 0.355$).

Total sweat loss and sweat rate

There was a main effect for testing sessions on total sweat loss ($P < 0.001$) and sweat rate ($P < 0.001$). Total sweat loss and sweat rate were not different between LI (mean $\pm$ SD: 1.2 ± 0.3 L, range: 0.7–1.7 L and 0.8 ± 0.2 L.h⁻¹, 0.5–1.1 L.h⁻¹ for sweat loss and sweat rate, respectively) and HI (1.4 ± 0.5 L, 0.4–2.1 L and 0.9 ± 0.3 L.h⁻¹, 0.3–1.4 L.h⁻¹, for sweat loss and sweat rate, respectively) training sessions when completed in a cool environment ($P = 0.411$ and $P = 0.397$ for total sweat loss and sweat rate, respectively). Total sweat loss and sweat rate were higher during HI warm (2.5 ± 0.4 L, 2.0–3.4 L and 1.7 ± 0.3 L.h⁻¹, 1.3–2.3 L.h⁻¹ for sweat loss and sweat rate, respectively) compared to HI cool (both $P = 0.001$, both $d = 1.62$). Total sweat loss and sweat rate were also higher during HI warm compared to LI cool (both $P < 0.001$, both $d = 1.78$). Mean and individual responses for sweat loss and sweat rate data can be found in Figure 1.

Fluid intake and body mass loss

There was a main effect for training sessions on fluid intake ($P < 0.001$). Fluid intake during the training sessions was not different ($P = 0.223$) between LI Cool (mean $\pm$ SD: 0.6 ± 0.1 L,

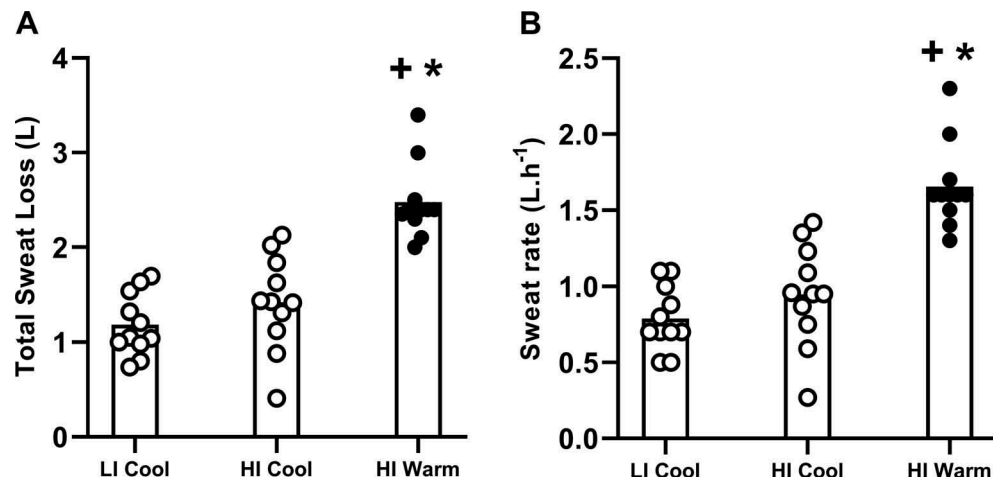


Figure 1. (a) Total sweat loss (L) and (b) Sweat rate ($\text{L}\cdot\text{h}^{-1}$) during the three testing sessions. 'LI' refers to low-intensity training load and 'HI' refers to high-intensity training load. Points represent individual values for each player in cool (white colour) and warm (black colour) condition, while bars represent mean values. * denotes a significant difference between the LI Cool testing session and the HI Warm testing session ($P < 0.001$); + denotes a significant difference between the HI Cool testing session and the HI Warm testing session ($P < 0.001$).

range: 0.4–0.8 L) and HI Cool (0.7 ± 0.1 L, 0.4–0.8 L). Fluid intake was higher ($P \leq 0.001$) during HI Warm (1.5 ± 0.2 L, 1.1–1.8 L) compared to HI Cool (0.7 ± 0.1 L, $d = 1.81$) and LI Cool (0.6 ± 0.1 L, $d = 1.91$). Despite *ad libitum* drinking made possible during training sessions, body mass was reduced after all sessions ($P = 0.033$) but in different proportions according to the environment (-1.0 ± 0.3 ; -0.7 ± 0.5 ; -0.6 ± 0.3 kg for HI warm, HI cool and LI cool, respectively). Body mass loss was more pronounced after HI warm than LI cool ($P = 0.015$, $d = 1.05$). However, no significant difference was found between LI cool and HI cool ($P = 0.760$) and between HI warm and HI cool ($P = 0.573$).

Sweat sodium concentration and sodium loss

A significant main effect ($P < 0.001$) of testing sessions was found for sweat sodium concentration (Figure 2). Sweat sodium concentration was lower during LI cool (mean $\pm$ SD: 16.1 ± 6.6 $\text{mmol}\cdot\text{L}^{-1}$, range: 6–30 $\text{mmol}\cdot\text{L}^{-1}$) compared to HI cool (54.6 ± 22.2 $\text{mmol}\cdot\text{L}^{-1}$,

22–83 $\text{mmol}\cdot\text{L}^{-1}$, $P < 0.001$, $d = 1.55$) and HI warm (66.5 ± 28.3 $\text{mmol}\cdot\text{L}^{-1}$, 28–106 $\text{mmol}\cdot\text{L}^{-1}$, $P = 0.001$, $d = 1.61$) sessions. Sweat sodium concentration was not different between the two HI training sessions completed in the cool and warm environments ($P = 0.263$).

There was also a significant main effect of testing sessions for total sodium loss ($P < 0.001$). Total sodium loss was lower during LI cool (mean $\pm$ SD: 779 ± 231 mg, range: 453–1100 mg) compared to HI cool (1405 ± 340 mg, 1005–1973 mg, $P < 0.001$, $d = 0.15$) and HI warm (2946 ± 958 mg, 1668–4500 mg, both $P < 0.001$, $d = 1.74$). Total sodium loss was higher during HI warm compared to HI cool ($P < 0.001$, $d = 1.51$).

Relationship between training load and hydration status

The relationship between total distance covered, distance covered at high-intensity, the number of high-intensity efforts, the number of accelerations and sweat loss during the three testing sessions is

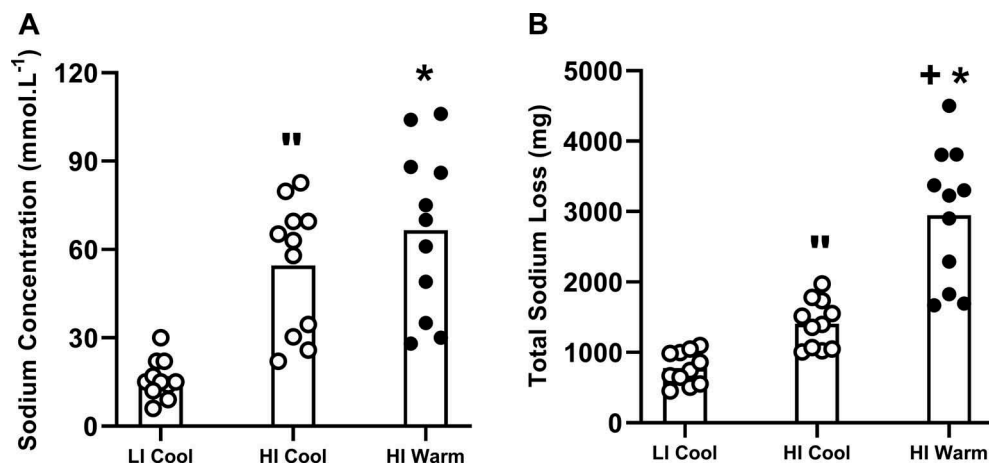


Figure 2. (a) Sweat sodium concentration ($\text{mmol}\cdot\text{L}^{-1}$) and (b) Total sodium loss (mg) during the testing sessions. 'LI' refers to low-intensity load and 'HI' refers to high-intensity load. Points represent individual values for each player in cool (white colour) and warm (black colour) condition, while bars represent mean values. * denotes a significant difference between the LI Cool testing session and the HI Warm testing session, + denotes a significant difference between the HI Cool testing session and the HI Warm testing session, " denotes a significant difference between the LI Cool testing session and the HI Cool testing session, where $P < 0.05$.

presented in Figure 3. There was a moderate correlation between total distance covered and sweat loss ($r = 0.48$, $P = 0.005$). Distance covered at high-intensity was not correlated to sweat loss ($r = 0.17$, $P = 0.341$). The number of high-intensity efforts ($r = 0.29$, $P = 0.104$) and intense accelerations ($r = 0.27$, $P = 0.248$) were not correlated to sweat loss.

The relationship between total distance covered, distance covered at HI, number of HI efforts, number of accelerations and sodium loss during the testing sessions is presented in Figure 4. A moderate correlation was found with sodium loss for the total distance covered ($r = 0.40$, $P = 0.023$) and the number of HI efforts ($r = 0.351$, $P = 0.045$). Distance covered

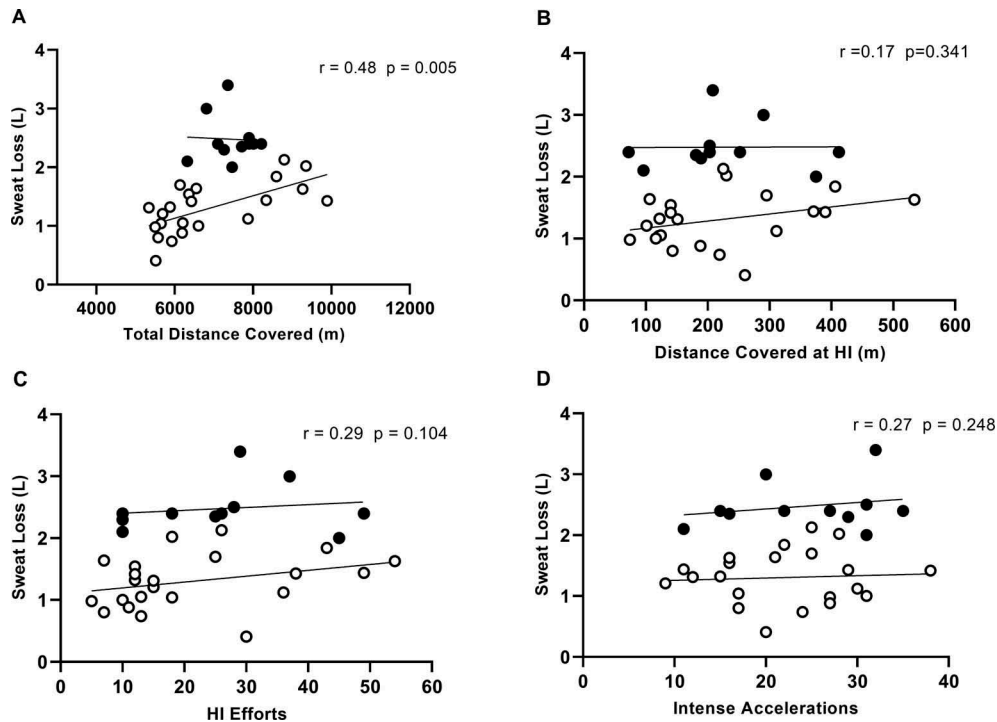


Figure 3. Relationship between (a) total distance covered (m), (b) distance covered at high intensity (m), (c) the number of high-intensity efforts, (d) the number of intense accelerations, and sweat loss (in L). Points represent individual values for each player in cool (white colour) and warm (black colour) conditions.

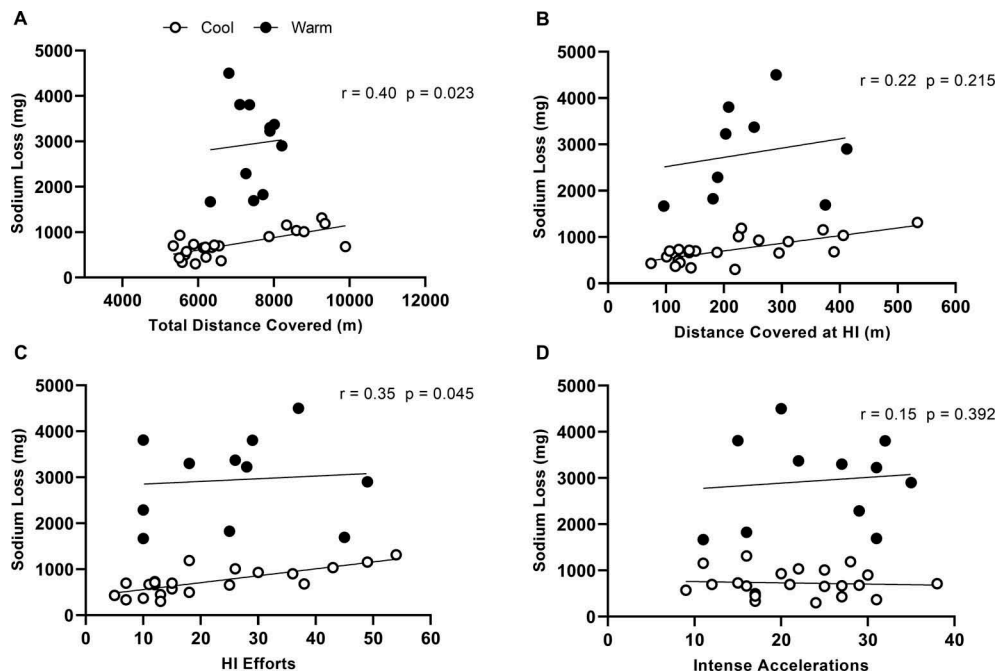


Figure 4. Relationship between (a) total distance covered, (b) distance covered at high intensity, (c) the number of high intensity efforts, the number of (d) of intense accelerations and sodium loss (mg). Points represent individual values for each player in cool (white colour) and warm (black colour) condition.

at HI ($r = 0.224$, $P = 0.215$) and the number of intense accelerations ($r = 0.150$, $P = 0.392$) were not correlated to sodium loss.

Discussion

The aim of this study was to examine the effects of training intensity and environmental condition on sweat rate, fluid intake, changes in body mass and sodium loss in elite football players of an English Championship League team. Our findings show that training intensity had no significant effect on total sweat rate and total sweat loss when training was performed in cool conditions. On the contrary, sweat rate, total sweat loss and total sodium loss were significantly higher when high intensity training was completed in a warmer environment. Regardless of training intensity and environmental condition, fluid intake did not match sweat loss in any of the training sessions. From the measured training load indices, the total distance covered on the pitch during the training sessions displayed the strongest correlation with both total sweat loss and sodium loss.

The effect of training intensity on hydration status

There is currently limited information on the effects of training intensity on sweat rate in football in different field environments. To our knowledge, only one study showed an intensity dependent relationship for sweat rate, with an increase of $0.4 \text{ L} \cdot \text{h}^{-1}$ between LI and HI training sessions, and a $0.2 \text{ L} \cdot \text{h}^{-1}$ increase in sweat rate between a HI training session and a simulated match (Duffield et al. 2012). It is noteworthy that in the latter study, all training sessions were completed in a warm environment. In our study, sweat rate also tended to increase in relation to training intensity with higher sweat rates recorded in the HI cool session compared to LI cool. However, this was not significant likely due to the large inter individual variability as sweat rate ranged from 0.5 to $1.1 \text{ L} \cdot \text{h}^{-1}$ in LI cool, and 0.3 to $1.4 \text{ L} \cdot \text{h}^{-1}$ in HI cool. Unsurprisingly, in the HI warm session, the sweat rate was exacerbated (1.3 to $2.3 \text{ L} \cdot \text{h}^{-1}$) compared to the other cool conditions. Consequently, total sweat loss was higher in HI warm (2.0 to 3.4 L) compared to HI cool (0.4 to 2.1 L) and LI cool (0.7 to 1.7 L) and in similar ranges to previous football studies conducted in either cool or warm environment (Maughan et al. 2004, 2005; Shirreffs et al. 2005). An increase in skin blood flow and adrenergic stimulation of sweat glands has been reported during high-intensity exercise, in response to the increase in core body temperature (Mora-Rodriguez et al. 2008). In our study, a significant positive relationship, albeit moderate ($r = 0.48$), was also found between total distance covered and total sweat loss, an outcome also observed by Duffield et al. (2012). However, no significant relationship was found between the indicators of high intensity (i.e. distance covered at high intensity, number of high intensity effort, and number of intense accelerations) in warm and cool conditions. The difference in intensity between LI and HI sessions was likely smaller and the ambient temperature was lower than in Duffield et al. (2012), thus likely limiting the increase in core temperature and the ensuing sweat production. However, an acknowledged limitation is that we did not monitor core temperature during the sessions, making the above explanations hypothetical.

Fluid intake and body mass loss were not different between the low and high intensity training sessions completed in a cool environment. The fluid intake values recorded in our study for the two cool training sessions match with those presented by Maughan et al. (2005) for football training in a cool environment. Fluid intake ranged from 0.4 to 0.8 L in LI and HI cool, and 1.1 to 1.8 L in HI warm, which correlates with variations in sweat rate and also confirms the variable nature of self-selected hydration strategies in football players (Maughan et al. 2004, 2005; Duffield et al. 2012). Our data also confirm that, regardless of training intensity, fluid intake rarely matched sweat rates, with only two players not incurring a reduction in body mass following the cool training sessions only. Body mass loss ranged from non-existent (0 kg ; 0% of body mass) to substantial (-1.8 kg ; -2% of body mass) across all sessions. The highest body mass loss (-2%) was identified during HI warm, however a 1.8% body mass loss was recorded for one of the players during HI cool. During the LI cool session, the highest level of dehydration recorded was 1.4% , while most players completed the training session with $< 1\%$ dehydration. Although a body mass loss $< 1\%$ may be considered as acceptable with limited impact on performance (Nielsen et al. 1982; Burke and Hawley 1997), these data indicate that special attention must be paid to hydration strategies when training load is increased (i.e. distance covered and or number of intensity bouts) even in a cool environment. More specifically, training conditions should be adapted to facilitate voluntary rehydration, including; i. providing breaks in training to allow for players to drink; ii. making the drinks easily available to the players; iii. paying attention to the organoleptic characteristics of the drinks (Hubbard et al. 1984).

Besides the loss of body water, sweating also induces electrolyte losses, with sodium being of particular interest as high sodium loss during exercise can result in important changes in plasma osmolality, and therefore accentuate dehydration and ensuing physiological perturbations (Shirreffs 2010). Sodium loss may alter muscle contraction and has been previously associated with exercise-induced muscle cramps in team sport athletes (Maughan et al. 2004; Stofan et al. 2005; Horswill et al. 2009). A training intensity dependent relationship has been reported for sweat sodium concentration and sodium loss in laboratory-based studies (Holmes et al. 2011; Baker et al. 2018). Our findings corroborate previous studies, showing that both sweat sodium concentration and sodium loss were significantly higher for all the players during HI cool ($+97 \pm 90\%$) compared to the LI cool session. A positive correlation was also found between sodium loss and total distance covered ($r = 0.40$) as well as between sodium loss and the number of HI efforts ($r = 0.35$). Even though these correlations were moderate and concerned only 11 players of 1 football team, these findings suggest that the total distance covered and the number of high-intensity drills completed during a training session should be considered for determining the quantity of electrolytes to be consumed during and after the training sessions. In a warm environment only, Duffield et al. (2012) added that total sodium loss might not be associated with intensity, but rather total distance covered. This is based on the fact that sweat rate cannot continue to rise with increases in core temperature when high intensity bouts are repeated, because there is no

linear relationship between skin blood flow and core temperature (Mora-Rodriguez et al. 2008). However, when core temperature is maintained elevated during the entire training session (at low or high intensity), resulting in a greater distance covered, sweat rate is maintained high and more sodium is lost.

The effect of environmental conditions on hydration status

Environmental conditions, particularly ambient temperature and humidity greatly influence sweat rate and electrolyte loss during exercise (Gleeson 1998). To the best of our knowledge, our study is one of the first to report the effects of different environmental conditions on indicators of hydration status in a field setting with similar physical load between training sessions. Our findings show a $0.8 \text{ L}\cdot\text{h}^{-1}$ mean increase in sweating rate during the HI warm session ($1.7 \pm 0.3 \text{ L}\cdot\text{h}^{-1}$) compared to the intensity matched training session completed in a cool environment ($0.9 \pm 0.3 \text{ L}\cdot\text{h}^{-1}$). Given the broad range of mean sweat rate ($0.3\text{--}2.5 \text{ L}\cdot\text{h}^{-1}$) in football reported in the literature, comparisons with previous studies is difficult (Nuccio et al. 2017). Many factors are susceptible to influence the sweat rate including the acclimation status, intrinsic characteristics, training characteristics, and the environment (Nuccio et al. 2017). In our study, as expected, fluid intake was higher during the training session completed in the warm environment compared to the cool environment ($1.5 \pm 0.2 \text{ L}$ vs $0.7 \pm 0.1 \text{ L}$), therefore, partially compensating for the higher sweat rate observed in the warm environment. Furthermore, 5 players incurred a body mass loss $\geq 1.5\%$ at the end of the HI warm session. Such level of dehydration may negatively impact exercise performance (Edwards et al. 2007).

Field-based data collected in various environments (from cold to hot) indicate a similar range of sodium concentration in the sweat lost during football exercises (Maughan et al. 2004, 2005; Kurdak et al. 2010; Duffield et al. 2012). This suggests that environmental conditions would not directly impact sweat sodium concentration. Nevertheless, although sweat sodium concentration was similar between these studies, total sodium loss was systematically higher in the studies carried out in warm and hot environments (Kurdak et al. 2010; Duffield et al. 2012) compared to cool environments (Maughan et al. 2005). Similarly, our findings showed no difference in sweat sodium concentration between the two HI training sessions completed in cool and warm conditions. However, total sodium loss increased by $101 \pm 95\%$ during the HI warm session compared to the HI cool session. Given that the sweat sodium concentrations were similar between the two sessions, the higher sodium loss recorded in the warm environment was likely the result of a significantly higher sweat loss (Sayer et al. 2019). Since total distance covered and intensity were not different between HI cool and HI warm, an increase in core temperature in HI warm likely increased skin blood flow and sweat rate. Our values for sodium loss across the sessions place within the range of values reported by Kurdak et al. (2010) in a warmer environment (34°C), but are higher than the values reported by Maughan et al. (2005) in a colder environment (5°C), therefore further emphasising the effect of environmental condition on sweat sodium loss. Taken together, our data and those from the literature

show a substantial variation in sweat sodium loss during training and match play, which prevents from defining clear guidelines for sodium requirements in football. However, a sweat sodium loss $<3000\text{--}4000 \text{ mg}$ during a training session or a match is often considered as reasonable and unlikely to alter performance (Coyle 2004). In our study, mean total sodium loss was $<1000 \text{ mg}$, $<1500 \text{ mg}$ and $<3000 \text{ mg}$ in LI cool, HI cool and HI warm, respectively. Only 1 player incurred a total sodium loss $>4000 \text{ mg}$ (4500 mg) in the HI warm session and thus would benefit from consuming a drink containing sodium during training sessions or matches performed in a warm environment.

Study limitations

Some limitations to this study need however to be acknowledged. Firstly, the final number of participants that took part in all training sessions was low ($n = 11$) and all the participants belonged to the same and only team. Although all the participants completed the same training sessions, GPS data shows slight inter-individual variability for several training indices (i.e. total distance covered, number of high-intensity efforts, distance covered at high intensity and number of accelerations). Accordingly, the intensity at which the training sessions were completed was slightly different between the players, particularly due to the high variability seen in high-intensity distance covered and number of accelerations, which could have affected the correlation between these training indices and total sweat loss and sodium loss. In all training sessions, players were allowed to drink *ad libitum* in order to evaluate whether their drinking strategy was adapted to the body water lost, hence reflecting real-world conditions. However, different drinking strategies between sessions and between players might have influenced data collected for sweat and sodium loss. The influence of ambient temperature on body temperature might have influenced the thermoregulatory mechanisms in addition to hypohydration. Lastly, the participants were not accustomed to training in warm conditions (they trained daily at a high latitude in the UK), therefore our results may not be generalised to soccer teams that are acclimatised to heat because they often train and compete in warm or hot environments. Further research is therefore required to elucidate the relationship between heat, hypohydration and dehydration in football. In future studies, fluid intake and training load should be standardised and body heat production measured for more accuracy. Despite its limitations, this study provides more information on the influence of training intensity and the environment on the hydration status of elite football players in a real-world applied setting. We believe these data are valuable for practitioners and players seeking to optimise hydration strategies in football.

Conclusion

We collected data during three separate football training sessions and showed that training intensity increased sweat sodium concentration and total sodium loss in elite football players of the same team. We also showed for the first time that warm environmental conditions increased sweat rate, fluid intake and total sodium loss compared to cooler conditions in

the same players under the same training load. Lastly, we showed that the total distance covered was correlated with total sweat loss and total sodium loss both in cool and warm environment. This data, albeit limited to one English Championship League team, offers real-world information that may orientate practitioners working with elite football players.

Practical implications

Our data confirm that sweat loss, sodium loss and fluid intake are highly variable between elite football players. Fluid intake rarely matched sweat rates, with only two players not incurring a reduction in body mass following training sessions across all testing conditions. In a cool environment (11–12°C), sweat rate varied from 0.5 to 1.1 L.h⁻¹ and reached up to 1.4 L.h⁻¹ during training sessions of low and high intensity, respectively. It seems that the total distance covered during training sessions is the most correlated to sweat rate. When the same high intensity session is performed in a warmer environment (23°C), sweat rate may reach up to 3.4 L.h⁻¹ and induce a substantial decrease in body mass (up to 2% in this study) when fluid intake is not sufficient. The risk of excessive sodium loss seems less likely since in our study only 1 player incurred a high sodium loss (>4000 mg in total) after the training session performed in the warm environment only. As such these findings suggest that sweat loss should be monitored in priority in any training condition (i.e. low and high intensity and in any environmental condition) to elaborate bespoke hydration strategies for the players. Sodium loss monitoring could be added in warmer conditions as the higher sweat rate may accentuate sweat sodium loss. In summary, the total distance covered during training session and the environmental condition in which the session takes place should be considered when providing hydration recommendations to football players.

Disclosure statement

No potential conflict of interest was reported by the authors.

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